

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		$\Delta y = 1.6 \text{ m[m]}$ or by implication (1) Correct use of $\lambda = \frac{a \Delta y}{D}$ ignoring slips of 10^n . ecf on Δy (1) $\lambda = 533 \text{ nm}$ [ecf on Δy if between 1.0 and 2.0 mm] (1)	1	1				
		(ii)		<i>Diffraction</i> mentioned (1) For each slit, intensity falls off with angle from normal or doesn't spread round through 180° or equivalent (1)		2		2		2
		(iii)	I.	$\lambda_{i-r} > \lambda_{vis}$ or $\lambda_{i-r} > 700 \text{ nm}$ or $\lambda_{i-r} > 533 \text{ nm}$ (1) Accept wavelength has increased or infra-red has a longer wavelength $\Delta y \propto \lambda$ or equivalent e.g. Δy increases because a and D are constant [so student is right] (1)		2		2		2
			II.	Increase D or decrease a		1		1	1	1
	(b)			Light is a wave / wave-like (1) Repetition of experiment by other scientist(s) (1) Other experiments completed to check on the nature of light (1)			3	3		
				Question 4 total	1	7	3	11	3	8